MIL-V-38398A(USAF)
8 July 1974
Superseding
MIL-V-38398(USAF)
16 February 1965

MILITARY SPECIFICATION

VALVE, STARTER CONTROL, PNEUMATIC, AIRCRAFT ENGINE GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for pneumatic control valves for use with the starters on aeronautical gas turbine engines.

1.2 Classification. Pneumatic starter control valves shall be one of the following two types:

Type I: Regulating (pressure).

Type II: Non-Regulating (on-off).

The military type and model designation will be assigned by the procuring activity in accordance with MIL-STD-875 and shall be specified in the model specification.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	Preservation-Packaging, Method Of
MIL-T-5021	Tests; Aircraft and Missile Welding, Operators' Qualification
MIL-B-5087	Bonding, Electrical, and Lightning Protection for Aerospace Systems
MIL-C-6021	Castings, Classification and Inspection of (For Aeronautical Applications)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

MIL-I-6866	Inspection, Penetrant Method of
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-E-7080	Electric Equipment, Aircraft, Selection and Installation of
MIL-P-7105	Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT, General Requirements for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series; General Specification for
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft and Missiles
MIL-S-8516	Sealing Compound, Polysulfide Rubber, Electric Connectors and Electrical Systems, Chemically Cured
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-D-8706	Data and Tests, Engineering, Contract Requirements for Aircraft Weapon Systems
MIL-S-8879	Screw Threads, Controlled Radius Root with Increased Minor Diameter, General Specification for
MIL-S-23586	Sealing Compound, Electrical, Silicone Rubber, Accelerator Required
MIL-I-81550	Insulating Compound, Electrical, Embedding, Reversion Resistant Silicone

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-453	Inspection, Radiographic
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-470	Maintainability Program Requirements (For Systems and Equipment)
MIL-STD-480	Configuration Control - Engineering Changes, Deviations and Waivers
MIL-STD-490	Specification Practices
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-875	Type Designation System for Aeronautical and Aeronautical Support Equipment
MIL-STD-889	Dissimilar Metals

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

MS 20995	Wire, Safety or Lock
MS 33540	Safety Wiring, and Cotter Pinning, General Practices for
MS 33588	Nut, Self-Locking, Aircraft Design and Usage Limitations of

PUBLICATIONS

Air Force - Navy Aeronautical Bulletin

438	Age Controls of Age-Sensitive Elastomeric Items

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Model specification and data submittal. When specified in the procurement contract a valve model specification conforming to the outline and instructions for preparation specified herein, shall be prepared by the contractor and submitted to the procuring activity for approval. (See 6.2 and 6.3.6). The model specification shall provide all the necessary information to completely describe the starter control valve for one weapon system application. Approval of the model specification is required prior to initiation of preproduction and quality conformance testing. Changes and revisions to the model specification shall be in accordance with MIL-STD-490.

a. The headings and numbering of sections and paragraphs in the model specification shall correspond to those in this specification. Specific data required by this specification and such other additional data as may be required to completely describe the unit shall be included in the model specification. Omission of reference in the model specification to a particular requirement of this specification by the manufacturer shall be interpreted as compliance therewith. When departures from the requirements of this specification are necessary, the details of such departures shall be stated as specific requirements bearing the same section, paragraph heading, and numbering as in this specification.

b. All applicable documents shall be listed in numerical order in Section 2 of the model specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

c. All curves shall be presented on graph paper having an adequate number of subdivisions to permit easy interpolation.

d. Definitions and symbols. Terms, their definitions, and symbols used in the model specification shall be in accordance with 6.3 herein.

3.2.1 Detail specification. Requirements for individual items shall be as specified herein and shall be in accordance with the applicable detail specification (see 6.3.7) for the system.

3.2.2 Valve data submittal and approval. Valve data submittal and approval requirements for a specific model aircraft will be covered by a contract with the appropriate procuring activity. Typical information and data required are listed in 6.2. The data shall be furnished in accordance with MIL-D-8706 or in accordance with the appropriate line items of the Contractor Data Requirement List (DD Form 1423), as applicable.

3.3 Mockup. A full-scale mockup shall be prepared for examination and aircraft installation suitability as soon as the contractor has established the necessary design features of the valve. Prior to initiation of the preproduction testing, the mockup shall be installed on the aircraft for which the design was finalized to demonstrate satisfactory fit and installation features. Examination shall be at the option of the procuring activity.

3.3.1 Changes. Changes to the valve features requiring changes in the vehicle or ground support equipment made after approval of the mockup shall be submitted to the procuring activity for approval in accordance with MIL-STD-480. Any changes required by the procuring activity shall be subject to negotiation as provided in the contract. Unless otherwise authorized, the mockup shall be kept current with approved changes at least through the first production contract.

3.4 Components. Normally, the complete valve shall consist of a valve body assembly which includes a mechanism (such as butterfly) for controlling air flow through the valve, a pressure regulating or on-off actuating power device, and a solenoid operated pilot valve for control of the actuator.

3.5 Selection of specifications and standards. Specifications and standards for necessary items and services not specified herein shall be selected according to MIL-STD-143.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

3.6 Materials

3.6.1 Metals. Metal parts shall be of the corrosion-resistant type or suitably treated to resist corrosion due to combustion gases or residue, or atmospheric conditions likely to be met in storage or service use. Any corrosion resistant protective coating applied shall provide maximum protection against cracking, chipping or scaling of the coating with age, use, or extremes of atmospheric conditions.

3.6.1.1 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.6.2 Nonexpendable metal parts. Nonexpendable metal parts may be reworked and reused as agreed between the procuring activity and the contractor, if such parts comply with specifications and drawings in all respects.

3.6.3 Synthetic rubber parts. Age of synthetic rubber parts shall be controlled in accordance with ANA Bulletin 438.

3.6.4 Castings. Castings shall be in accordance with MIL-C-6021.

3.6.5 Potting compounds. Requirement 47 of MIL-STD-454 shall be adhered to for encapsulation and embedment. Only those materials which meet the requirements of MIL-S-8516, MIL-S-23586, or MIL-I-81550 shall be used.

3.6.6 Organic materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used, they shall be treated with a suitable fungicidal agent.

3.7 Design and construction. The valve shall be designed for operational use with a pneumatic energy source, and its design shall be amendable to production processing methods. The valve shall be constructed to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service usage. The pneumatic energy sources shall be defined in the detail specification and model specification.

3.7.1 Special features

3.7.1.1 Failsafe features. The valve shall be designed to automatically close in the event of loss of electrical power or pneumatic energy source.

3.7.1.2 Other features. Manual override, remote position indication, and other such features shall be specified in the detail specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

3.7.2 Reliability and maintainability

3.7.2.1 Reliability. The valve manufacturer shall prepare, as part of the model specification, a reliability analysis predicting the confidence level and reliability factor based on a one-cycle mission. Demonstration of the predicted reliability factors shall be specified in the contract with the procuring activity. Factors to be considered during the design and manufacture of the valve to assure in-service reliability are:

a. Simplicity of design.

b. Selection and application of reliable components.

c. Consideration of operational and environmental parameters.

d. Mechanical structures - adequate design stress margin on all parts.

e. Design review by technical specialists prior to drawing release.

f. Use of proven manufacturing techniques.

g. Rigid quality control procedures imposed throughout procurement, manufacturing, assembly, and testing to assure the maximum design reliability is built into the equipment.

3.7.2.2 Maintainability. The valve design shall permit easy assembly, disassembly, location of trouble sources, routine service checking, adjustment, and maintenance with tools and equipment normally available commercially, by service maintenance personnel with a minimum of training. Maintenance between overhaul periods shall be held to an absolute minimum.

3.7.3 Flight and ground loading conditions. The valve shall withstand, without permanent deformation or failure, the largest forces resulting from ground loading conditions and the largest forces resulting from all critical combinations of maximum flight maneuver conditions specified in the detail specification, acting in any direction through the center of gravity of the valve. In addition, unless otherwise specified, the valve's air inlet connecting flanges shall be capable of withstanding, without permanent deformation or failure, a force of 222N (50 pounds) direct thrust acting either inward or outward or a moment of 169NM (150 inch pounds) acting in any direction around the axial centerline of the flange.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

3.7.4 Threads

3.7.4.1 Straight screw threads. All conventional straight screw threads shall conform to the requirements of MIL-S-8879 Classes 3A or 3B. The use of MIL-S-7742 is optional for threads used for:

a. Electrical connections.

b. Screw threads 3.5 mm (.138 inch) diameter and smaller.

c. Interference fits and other applications where MIL-S-7742 threads are suitable for the intended purpose, such as installation of studs or external threads of inserts and their mating tapped holes.

3.7.4.2 Tapered pipe threads. Tapered pipe threads shall be in accordance with MIL-P-7105 and may be used only for permanently plugging drilled or cored openings.

3.7.4.3 Threaded inserts. All threaded connections in nonferrous materials which are heavily loaded or removed frequently shall have steel inserts which are suitably protected from electrolytic corrosion.

3.7.5 Inspection seals. Calibration adjustment points and other critical points shall be assembled at the factory with inspection seals so that no part of the valve can be disassembled without breaking the seal.

3.7.6 Safetying. All threaded connections shall be locked to prevent loosening in service by means of safety wiring, by self-locking nuts, cotter pins, or other methods approved by the procuring activity. Safety wire shall conform to MS20995 and be installed in accordance with MS33540. The use of lockwashers or staking is prohibited if used as a primary locking device. The general design and usage limitation of self-locking nuts specified in MS33588 shall be applied.

3.7.7 Drainage. The valve shall incorporate provisions for drainage or be sealed to prevent water from accumulating within the unit. Drainage shall be such that any condensed or accumulated water will not cause malfunction or cause delay in operation if frozen. Details of valve orientation for the aircraft installation shall be included in the detail specification.

3.7.8 Direction of flow. The direction of air flow through the valve shall be prominently and permanently marked on the valve housing.

3.7.9 Snap ring retainer. The use of snap ring retainers is prohibited except where failure of the retainer will not affect operation of the valve.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

3.7.10 Pneumatic bleed. Bleed ports and passages shall be as large as is practical to preclude blockage by contamination.

3.7.10.1 Continuous bleed. If the valve incorporates any requirement for continuous bleed, it shall be considered as a part of the total valve leakage as defined in 3.9.4 and shall be approved by the procuring activity.

3.7.10.2 Filter. If filters are required, the maximum opening shall be sufficient to protect the valve from the contamination required in 4.6.13.2. The filter shall be of sufficient size to minimize the need for cleaning and shall be readily removable for replacement or cleaning without performing a major overhaul on the valve.

3.7.11 Envelope requirements. All components of the valve shall be contained within the envelope and the mounting flanges and electrical connector shall be as defined in the detail specification. The valve shall meet all of the applicable requirements of this specification when mounted in any position. Dimensions and tolerances not specified shall be as close as is consistent with high quality shop practices. Where dimensions and tolerances may affect interchangeability, durability, or performance of the valve, they shall be limited and held accordingly.

3.7.12 Electrical requirements. The valve shall meet the requirements of this specification with electrical power from a 28 volt dc system conforming to MIL-STD-704. All electrical equipment shall be designed in accordance with MIL-E-7080. Steady state operating or holding current of the unit shall be specified in the detailed specification.

3.7.12.1 Solenoid preload. The solenoid armature and pilot valve parts shall be designed to provide sufficient preload in the closed position to overcome the forces associated with the environmental conditions (vibrations, etc.) specified in the detail specification. If pressure assist is utilized in accomplishing this, the conditions must be met with the minimum operating pressure specified in the detail specification.

3.7.12.2 Connectors. Electrical connectors shall be in accordance with the detail specification. Provisions for lockwiring the connector plug shall be incorporated at the electrical receptacle.

3.7.12.3 Dielectric strength. All parts of electrical equipment, except critical components such as transistors and capacitors, shall be capable of withstanding, at commercial frequency, a voltage of 1,000V (rms) plus twice the working voltage. Transistors and capacitors shall withstand twice the peak voltage to which they will be subjected during service or 100V, whichever is greater, for a period of 1 minute.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

3.7.13 Bonding. Bonding control shall be in accordance with the requirements of MIL-B-5087.

3.7.14 Lubrication. The valve shall not require relubrication during the overhaul life.

3.7.15 Position indicator. The valve shall be equipped with an external position indicator which shall include suitable markings to indicate the valve (open) and (closed) positions.

3.7.16 Proof and minimum burst pressure. The valve shall meet the operational requirements of this specification and shall not have evidence of damage or permanent deformation after having been subjected to a proof pressure equal to two times the maximum operating pressure specified in the detail specification applied at the inlet port with the valve in both the open and closed positions and while at maximum operating temperatures. Minimum burst pressure shall be one and one-half times the proof pressure.

3.7.17 Weight. The actual weight of the complete unit shall be specified in the model specification and shall not exceed the maximum weight specified in the detail specification.

3.7.18 Design and fabrication changes. No changes shall be made in the design of parts or materials listed in the approved valve parts list except when such changes are approved in accordance with the provisions of MIL-STD-480. No changes shall be made in vendors or fabrication sources except when such changes are approved by the procuring activity. Approval of changes does not relieve the contractor of full responsibility for the results of such changes on valve characteristics.

3.7.19 Parts list. The parts list for the valve that successfully completes the preproduction test shall constitute the approved parts list for subsequent valves of the same model to be delivered to the procuring activity.

3.8 Inspection process and certification

3.8.1 Magnetic particle inspection. Magnetic particle inspection shall be in accordance with MIL-I-6868.

3.8.2 Fluorescent penetrant inspection. Fluorescent penetrant inspection shall be in accordance with MIL-I-6866.

3.8.3 Fusion welding certification. All operators performing fusion welding shall be certified in accordance with MIL-T-5021.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

3.8.4 Radiographic inspection. Radiographic inspection shall be in accordance with MIL-STD-453.

3.8.5 Hydrostatic test methods. Hydrostatic test methods shall be approved by the procuring activity.

3.9 Performance characteristics. The valve performance characteristics shall be specified in the model specification. The valve after being subjected to the tests specified in section 4 shall meet the operating requirements specified herein. All performance characteristics shall be based upon the use of production hardware.

3.9.1 Valve operation. The valve shall be normally closed with the inlet side of the valve pressurized to any inlet condition as specified in the detail specification. Application of electrical power to the unit shall cause the valve to open. The valve shall remain open as long as pressure in maintained and the electrical circuit is energized.

3.9.1.1 Opening-Closing time. With the inlet side of the valve pressurized to any inlet condition specified in the detail specification, and when the electrical circuit to the valve is energized, the valve shall move from full closed position to the required regulating pressure if the valve is the regulating type, or to the full open position for the nonregulating type. With the valve inlet pressurized to the maximum pressure inlet condition, the rate of pressure rise shall not exceed that specified in the detail specification. Unless otherwise authorized by the procuring activity, the valve shall not take longer than 2 seconds from time of electrical actuation to reach the required open position at any ambient or condition of pressure and temperature. When the electrical circuit is interrupted, the valve shall close in not less than .4 seconds nor more than 2 seconds under any condition.

3.9.1.2 Regulation. Pressure regulating valves shall meet the opening-closing time requirements specified in 3.9.1.1 and shall regulate to the pressure specified in the detail specification.

3.9.1.3 Breakaway torque. Available breakaway torque of the flow controlling mechanism at normal operating pressure shall be at least 20 percent higher than that required, after accounting for the most severe mechanism icing condition that could be encountered in service.

3.9.2 Duty cycles. The valve shall be capable of continuous cycles of 60 seconds open followed by 60 seconds closed and also 5 minutes open followed by 5 minutes closed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

3.9.3 Pressure drop. Unless otherwise specified in the detail specification, the pressure drop from inlet to outlet with air flowing at rated inlet conditions through a non-regulating type valve, or at rated inlet conditions with controlling mechanism held open through a regulating type valve, shall not exceed .02 Bar (0.3 psi).

3.9.4 Leakage. With the valve in the closed position and with upstream pressure and temperature at any inlet condition specified in the detail specification, internal air leakage from the inlet port to the outlet port shall not exceed .09 Kg per minute (0.20 pounds per minute) and total leakage (internal plus external) shall not exceed .18 Kg per minute (0.40 pounds per minute).

3.9.5 Endurance. The valve shall be capable of sustaining 10,000 cycles without overhaul or replacement of parts, and shall operate within the limits specified in the detail specification when subjected to the Cycling and Endurance Test specified under 4.6.8.

3.9.6 Explosion proof. The valve shall be made explosion proof (it shall not cause ignition of an ambient explosive gaseous mixture).

3.9.7 Temperature extremes. The valve shall meet the requirements of this specification when subjected to the extreme thermal conditions of its installation that are associated with the following temperature conditions:

a. Ambient temperature operating range -54°C to +149°C (-65°F to +300°F).

b. Exposure temperature range -73°C to 204°C (-100°F to +400°F).

3.9.8 Performance curves. The valve performance shall be defined by the following curves which shall be a part of the model specification:

a. Curves showing maximum pressure rise versus time at each specified inlet condition at sea level.

b. For pressure regulating valves, curves showing maximum peak pressures and regulated pressures (with allowable tolerance limits) for each specified condition of inlet pressure greater than the regulated pressure.

c. Curves showing variations in pressure rise time and regulated pressure with altitude and temperature.

3.9.9 Compatibility. The procuring activity may require that the control valve be compatible with a starter specified in the detail specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

3.10 Environmental conditions. The valve shall suffer no detrimental effects and shall operate satisfactorily during and after exposure to the natural and induced environments including altitude, temperature, temperature shock, rain, salt, fog, dust, fungus, humidity, vibration, electromagnetic interference, shock, and other environmental conditions as anticipated in service.

3.10.1 Field operation. The valve shall operate within the limits specified in the model specification under field operating conditions.

3.11 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. Matched parts or selective fits will be avoided wherever possible; however, where required these parts shall be serialized. The drawing number requirements of MIL-STD-100 shall govern changes in the manufacturer's part numbers. Physical interchangeability shall be in accordance with MIL-I-8500.

3.12 Finishes. All aluminum parts shall be anodized in accordance with MIL-A-8625. Unless otherwise specified by the procuring activity, all exposed surfaces except aluminum parts, corrosion-resisting steel, working surfaces, threads, and plated parts shall be finished with one coat of primer followed by two coats of black enamel.

3.13 Identification of product. The valve shall be marked for identification in accordance with MIL-STD-130 as follows:

Nomenclature
Manufacturer's Part Number
Manufacturer's Serial Number
Federal Stock Number
Contract Number

3.14 Workmanship. The workmanship and finish shall be of sufficiently high grade to insure satisfactory operation, reliability, and durability consistent with the application and storage life requirements of the valve.

3.15 Deviations. The contractor shall include in the model specification under this paragraph a complete listing of all deviations requested to the requirements of this specification and the detail specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 All tests shall be subject to witnessing by an authorized representative or representatives of the procuring activity.

4.2 Classification of tests. The inspection and testing of valves shall be classified as follows:

a. Preproduction tests.

b. Quality conformance tests.

4.3 Test conditions. Unless otherwise specified herein, the following conditions shall apply for the valve tests.

4.3.1 Temperature. Ambient temperature is defined as the existing temperature within the test cell at time of test. Valve conditioned temperatures shall be the temperature of the valve after it is soaked. Low temperature tests shall be conducted at -53.9°C (-65°F).

4.3.2 Pressure. Unless otherwise specified herein, all tests shall be conducted at ambient atmospheric pressure.

4.3.3 Inlet condition. Unless otherwise specified, all valve cycling tests shall be conducted at rated inlet conditions. The pressure and temperature parameters for the inlet conditions not specified herein shall be specified in the detail specification for the applicable valve installation and may include the following:

a. Rated.

b. Maximum temperature.

c. Minimum pressure.

d. Maximum pressure.

e. Maximum engine interbleed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

4.3.4 Conditioning time. Conditioning time for the valve shall be such that all parts of the valve shall have reached a temperature within the specified temperature limits. During the conditioning time, the temperature of the conditioning chamber shall not vary more than ±2.8°C (±5°F) from the specified temperature.

4.3.4.1 Long term low temperature soak. Soak the valve until temperature stabilization is reached.

4.3.5 Transfer time. The valve transfer time shall not exceed 5 minutes.

4.3.6 Test apparatus

4.3.6.1 Accuracy. For all valve tests the test apparatus shall be such as to insure that recorded parameters will have the accuracy specified in MIL-STD-810. Other parameters will have a steady-state accuracy within ±2 percent except that the weight shall be accurate within ±0.2 percent. All apparatus shall be calibrated often to insure that this degree of accuracy is maintained. Calibration records shall be retained by the testing agency for 2 years and furnished to the procuring activity upon request.

4.3.6.2 Recalibration. The point at which an unusual difference occurs in any parameter shall be recorded and the valve shall be recalibrated. If this point occurs prior to the last cycle of testing, the test shall be stopped until the procuring activity is notified and gives approval to proceed.

4.3.7 Valve attitude during testing. All valve tests shall be conducted with the valve mounted in the most critical operational attitude. The mounting attitude shall be described in the model specification.

4.3.8 Test installations. Test installations for critical tests, such as pressure rise rate, shall be described in the detail specification.

4.3.9 Tests and inspections for preproduction and quality conformance approval. Tests and inspections required for the valve for the purpose of establishing preproduction approval and quality conformance approval will be limited to those tests specified herein except as modified for a particular valve design. All verification of requirements by means of results of testing on similar valves or components shall require approval of the procuring activity. These limitations of tests and inspections do not relieve the contractor of responsibility for fulfilling all requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

4.4 Preproduction testing

4.4.1 Preproduction test method. When specified (see 6.2), a preproduction test method shall be prepared. The test method shall have been approved by the procuring activity prior to the initiation of preproduction testing. This document shall not be a part of the model specification; however, any deviation to MIL-V-38398 shall be specified in the model specification.

4.4.2 Preproduction tests. Preproduction tests as proposed by the contractor in accordance with 4.4.1 shall be designed to demonstrate complete compliance with all the requirements of Section 3 herein. The following test schedule is provided as an outline for the tests to be conducted on each of the three test units. Specific tests required to demonstrate compliance with each requirement shall be integrated into the individual unit test schedules.

4.4.2.1 Test sample number one:

Test Paragraph	Test Title
4.6.7	Calibration
4.6.8	Cycling and Endurance
4.6.9	Proof Pressure Test
4.6.10	Vibration
4.6.11	Recalibration
4.6.12	Teardown and Inspection

4.4.2.2 Test sample number two:

4.6.7	Calibration
4.6.13	Dust
4.6.14	Rain
4.6.15	Low Temperature
4.6.24	Icing
4.6.16	Fungus (if applicable)
4.6.17	Salt Fog
4.6.18	Humidity
4.6.19	High Temperature
4.6.11	Recalibration
4.6.12	Teardown and Inspection

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

4.4.2.3 Test sample number three: The following tests may be conducted on one of the first two valves after completion of that valve's particular tests, or on a separate unit or separate components as applicable. These tests may be conducted in any sequence desired.

Test Paragraph	Test Title
4.6.2	Dielectric
4.6.21	Pressure Drop
4.6.22	External Loads
4.6.23	Burst Pressure
4.6.25	Explosion Proof

4.4.3 Preproduction test samples. A minimum of three valves are required as preproduction samples and shall be referred to hereafter as units one, two, and three. These test valves shall have been manufactured from production drawings and shall be identical to the proposed production valves. Maintenance, adjustment, or replacement of components or parts shall not be permitted after initial acceptance of the valves. Disassembly of any part of the valve to any extent prior to the final teardown inspection shall not be permitted.

4.4.4 Preproduction failure/success criteria. Preproduction shall be considered successful when all preproduction requirements of section 3 are satisfactorily met. When conditions exist such that an unqualified success cannot be assigned to a particular test the following applies.

a. When the success of a particular test cannot be determined due to instrumentation or a failure not related to the valve, that test shall be repeated at least once.

b. When a valve or valve component related failure or malfunction occurs during a test it shall be repeated at least twice.

4.4.5 Design changes during preproduction. If the design is changed after start of preproduction testing, tests associated with those performance characteristics that could be affected by design changes shall be rerun.

4.4.6 Notification of test failures. A failure analysis report shall be submitted to the procuring activity immediately after any failure of a preproduction test.

4.5 Quality conformance tests. The quality conformance tests shall consist of:

a. Individual tests.

b. Sampling plans and tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

4.5.1 Individual tests. Each valve shall be subjected to the following individual tests specified under 4.6 as a minimum. Additional tests may be required depending upon the particular valve design:

a. Examination of product (See 4.6.1).

b. Dielectric strength test (See 4.6.2).

c. Proof pressure test (See 4.6.3).

d. Cycling test (See 4.6.4).

e. Actuation test (See 4.6.5).

f. Leakage test (See 4.6.7).

4.5.2 Sampling plans and tests

4.5.2.1 Sampling plan A. Each valve of the first 10 production units, every tenth of the next 90, and every fiftieth production configuration valve thereafter shall be subjected to individual tests under 4.5.1.

4.5.2.2 Sampling plan B. The first and every fiftieth production unit thereafter shall be subjected to the following tests:

a. Examination of product (see 4.6.1).

b. Cycling test (see 4.6.4).

c. Actuation test (see 4.6.5).

d. Leakage test (see 4.6.7).

4.5.3 Rejection and retest. Failure of any unit to pass any individual test listed in 4.5.1 shall be cause for rejection of that unit (rework and retesting is acceptable). Failure of any sampling test on a unit from a lot shall be cause for rejection of the entire lot. Once a lot has been rejected by a procuring agency (government or industrial), full particulars concerning the cause of rejection and the action(s) taken to correct the defects in the lot shall be furnished in writing before the lot can be retested.

4.5.4 Quality conformance test method. A quality conformance test method, including performance parameters, shall be included as a part of the model specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

4.6 Test methods

4.6.1 Examination of product. Each valve shall be subjected to a careful inspection to insure proper assembly, configuration, workmanship, materials, finishes, processing, weight, identification, and compliance with applicable specifications.

4.6.2 Dielectric strength test. The electrical equipment shall be subjected to a test voltage of 1,000V (RMS) plus twice the working voltage at commercial frequency for a period of 1 minute. The test voltage shall be applied between the terminals (shorted together) and ground at sea-level atmospheric pressure. If the preceding method of testing is not feasible, the dielectric tests may be conducted on the components prior to final assembly or with the critical components disconnected. The test voltage for critical components shall be as specified in 3.7.12.3.

4.6.3 Proof pressure test. With the valve in the closed position and the outlet port open to atmospheric pressure, apply the proof air pressure specified in 3.7.16 to the inlet port of the valve for a period of one (1) minute. Repeat this test, except the valve shall be in the open position and the outlet port shall be capped.

4.6.4 Cycling test. The valve shall be operated for 25 cycles using air at rated inlet conditions at room temperature and 21 volts dc electrical power.

4.6.5 Actuation test. The valve shall be connected to a 28 volt dc electrical power source and actuated twice at each of the inlet conditions specified in the detail specification. Flow rate may be reduced to a valve where air temperature can be maintained, provided the opening rate is not affected. Valve operation shall be smooth and the opening and closing rates shall be within the limits specified herein and in the detail specification. Regulated pressure, if applicable, shall be within the limits specified in the detail specification. Current draw shall not exceed that specified herein.

4.6.6 Leakage test. Internal and total leakage shall be measured with the valve in the closed position and upstream pressure at both the maximum and minimum inlet pressures and temperatures specified in the detail specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

4.6.7 Calibration. The valve shall be subjected to four cycles at each of the inlet conditions specified in 4.3.3. Two cycles at each condition shall be conducted using 21 volts dc and two cycles using 28 volts dc. Inlet pressure and temperature limits shall be ±.07 Bar (±1 psia) and ±8.3°C (±15°F). Airflow through the valve shall be controlled by an orifice or nozzle downstream as defined in the detail specification. Individual test valves shall be rejected if performance is not within the limits defined in the detail specification. Data in accordance with 6.2.1.4 shall be recorded for each calibration cycle. In addition, it shall be demonstrated that the valve will open (timing requirements need not be met) with rated inlet conditions and 17 volt dc power. The initial calibration shall be conducted after the break-in run and before initiation of the scheduled preproduction test program. The starter shall not be operated between completion of the initial calibration runs and start of the scheduled test program.

4.6.8 Cycling and endurance test. Including the calibration cycles, the valve shall be subjected to a total of 10,000 endurance cycles at the conditions specified in table I. A cycle in this case is defined as actuation from close to open to close while the unit is exposed to the operational environmental conditions defined by this specification. Airflow through the valve shall be controlled as specified in 4.6.7 and electrical power shall be 28 volts dc. The valve shall remain in the open and closed positions for a minimum of 5 seconds during each cycle throughout this test, except that during period II the cycling rate shall be 60 seconds open, followed by 60 seconds closed continuously, and during period III the cycling rate shall be 5 minutes open followed by 5 minutes closed continuously. Data in accordance with 6.2.1.4 shall be recorded for the first, last, and every 500th cycle during each period of testing to show compliance with the opening and closing time requirements of this specification and internal and total leakage shall be measured after every period. Failure of the valve to meet the operation and leakage requirements throughout the test shall be cause for rejection.

TABLE I

PERIOD	NO. OF CYCLES	INLET CONDITION (1)	AMBIENT TEMPERATURE
I	2480	Maximum Pressure	Ambient
II	900	Maximum Temperature	High Operating
III	100	Maximum Temperature	High Operating
IV	5000	Rated	Ambient
V	1500	Maximum Temperature	High Operating

(1) The related inlet pressure or temperature conditions to those listed shall be specified in the detail specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

4.6.9 Proof pressure test. With the valve in the closed position and the outlet port open to atmospheric pressure, apply the proof air pressure specified in 3.7.16, to the inlet port of the valve for a period of 10 minutes. Repeat this test, except the valve shall be in the open position and the outlet port shall be capped. Following these tests, the valve shall function properly, and leakage shall not exceed the values specified in 3.9.4.

4.6.10 Vibration. Vibration tests shall be conducted in accordance with the following method and procedure:

4.6.10.1 Method. The valve used in the Vibration Test shall be the Endurance Test valve and shall be tested as a complete assembly. Before and after this test, a calibration shall indicate that no component subject to calibration has changed its calibration beyond the design tolerance range. All components not subject to calibration shall be operated and shall not have had any part of their function impaired by the test.

4.6.10.2 Procedure. The following procedure shall apply.

4.6.10.2.1 Test installation. The valve shall be mounted on a test fixture to simulate airplane installed position. If the aircraft installation requires that the starter control valve be mounted directly on the starter pneumatic inlet flange, the vibration shall be conducted with the applicable control valve mounted on a test fixture simulating the starter as in the airplane installation. To more closely simulate the airplane installation, the airplane pneumatic duct may be attached to the valve and supported as in the airplane. If the pneumatic ducting is simulated the test setup shall be coordinated and approved by the procuring activity. If the starter control valve is mounted remotely in the airplane installation, the valve shall be vibrated as a separate unit. The test fixture shall be rigid as possible and designed to eliminate or minimize fixture resonances in the test frequency range. The test configuration shall be such that rotational motion of the fixture is not induced due to any unsymmetrical weight or stiffness distribution of the equipment. Measurements of vibratory accelerations or displacement shall be made at the mounting base of the valve. All frequency measurements shall be accurate within $\pm$5 percent. All displacement or acceleration measurements shall be accurate within $\pm$10 percent. The motion of the vibration table shall be simple harmonic motion with not more than 20 percent distortion.

4.6.10.2.2 Frequency scan. The equipment shall be vibrated with essentially simple harmonic motion in each of three mutually perpendicular directions to explore for resonant frequencies of component parts over the ranges of frequencies required in the detail specification. Frequency scans shall be conducted very slowly and carefully. The frequency range shall be broken into small convenient intervals and each interval scanned at a constant applied force amplitude which produces approximately the fixture displacement or acceleration

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

amplitude defined in the detail specification. The fixture motion shall be observed closely during the frequency scans to detect the frequencies of minimum fixture motion which defines the frequencies at which major components of the unit are in resonance. The frequencies are quite sharply tuned and do not necessarily coincide with the frequencies at which maximum component amplitudes or noises occur when scanning at constant applied force amplitude. When the resonant components are small, the reduction of fixture motion at resonance may not be discernible, in which case the resonant frequencies may be determined from observations of maximum component amplitudes or noises, phase shift between fixture motion, and motion of a resonant part. In all cases, the resonant frequencies shall be verified, if possible, by checking for minimum fixture motion. The test report must contain complete descriptions of all resonances, including the resonant frequencies, parts in resonance and modes of vibration.

4.6.10.2.3 Resonance endurance tests. After completion of the frequency scan, resonance endurance tests shall be conducted at all resonant frequencies of the components, with the applied force adjusted as necessary to obtain the applicable fixture displacements or accelerations specified in the detail specification. The valve shall be continuously pressurized using air at a pressure equal to maximum engine interbleed pressure and shall be cycles five times during vibration at each resonant frequency. (Flow rate optional.) Operation of the valve shall be normal and any tendency for the valve to open without being actuated shall be cause for rejection. The vibrator and power supply must produce the prescribed fixture motions at all frequencies. Endurance tests shall not be conducted at any frequency at which the fixture amplitude abruptly increases when scanning at a constant applied force amplitude. Endurance tests shall be interrupted periodically to re-scan for resonance. If a change in resonant frequency occurs during endurance testing at resonance, the frequency of vibration shall be adjusted to follow the resonance. However, if large or abrupt resonant frequency shifts occur, the item shall be examined for component failures. Resonance endurance tests shall be conducted for one million cycles or eight hours, whichever occurs first, at each resonance. (See note I.) If the same resonance is excited by vibration applied in more than one direction, the endurance test for that resonance shall be accomplished with vibration applied in the direction which produces the greatest response. If no resonances are found, the unit shall be vibrated at 100 cps and the corresponding amplitude as specified in the detail specification for one million cycles in each of the three mutually perpendicular directions. (Note I: Where several resonances are encountered, the procuring activity shall be notified of the parts in resonance and, at the option of the procuring activity, the endurance tests may be limited to those resonances considered most likely to produce failure. A detailed description of each resonance is required.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

4.6.10.2.4 Cycling endurance test. The unit shall be subjected to the following cycling tests at room temperatures in addition to the frequency scan and resonance endurance tests. The unit shall be mounted as specified in 4.6.10.2.1 and shall be vibrated at the displacements and accelerations defined in the detail specification with the frequency continuously varied between the specified limits in a cycle of approximately 15 minutes. The rate of change of frequency shall be proportional to the frequency. (The rate of change of frequency may be a linear function with time.) The total test time shall be as specified in MIL-STD-810, method 514, procedure I. This test shall be repeated with vibration along each of three mutually perpendicular axes. The cycling test may be broken into convenient frequency ranges, provided the cycling rates and test times for each range are not changed.

4.6.11 Recalibration. The valve shall be subjected to a calibration as specified in 4.6.7. Failure of the unit to meet the performance requirements of the detail specification shall be cause for rejection.

4.6.12 Teardown and inspection. The valve shall be completely disassembled for examination of all parts and for measurements as necessary to disclose excessively worn, distorted, or damaged parts. These measurements shall be compared with those taken prior to the start of the Preproduction Test. The Preproduction Test shall be considered complete when every component of the valve has been subject to, and has completed, the specified amount of preproduction testing without damage to an extent which would prohibit further operation of the unit. Photographs of the parts as disassembly shall be made and furnished as a part of the Preproduction Test Report. Photographs of small parts and any wear points or failures shall be enlarged to show detail.

4.6.13 Dust test. The valve shall be subjected to Dust Tests in both operating and non-operating conditions. The valve shall be mounted in a manner similar to its intended installation. Data in accordance with 6.2.1.4 shall be recorded both before and after each part of the test. Leakage shall be measured both before and after the test. All cycles shall be conducted using rated inlet conditions for non-regulating valves and maximum engine interbleed conditions for regulating valves. Failure of the valve to meet the performance requirements of the detail specification shall be cause for rejection.

4.6.13.1 Part I, non-operating. The valve shall be subjected to a Dust Test in accordance with Method 510, Procedure I, of MIL-STD-810 while the valve is inoperative.

4.6.13.2 Part II, operating. The valve shall be cycled 50 times while being supplied with contaminated air to the valve in accordance with Method 510, Procedure I, of MIL-STD-810, except the air shall be supplied at either rated or maximum engine interbleed conditions as specified above, for the applicable type of valve.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

4.6.14 Rain test. The valve shall be subjected to a Rain Test in accordance with Method 506, Procedure I, of MIL-STD-810. Data in accordance with 6.2.1.4 shall be recorded for the test cycles both before and after the test. Rated inlet conditions shall be used for the test cycles.

4.6.15 Low temperature test. The valve shall be subjected to a Low Temperature Test in accordance with the following procedure:

a. Soak the valve at -53.9° ±2.8°C (-65° ±5°F) until stabilized at -53.9° ±2.8°C (-65° ±5°F).

b. While being maintained at the -53.9°C (-65°F) ambient, operate the valve for one cycle using rated inlet conditions.

c. Soak the valve at -53.9° ±2.8°C (-65° ±5°F) for 30 minutes and repeat the procedure in b above.

d. Repeat the procedure in c above.

e. The complete procedure above, with the exception that the initial soak may be reduced to 5 hours, shall be repeated 2 times making a total of 6 cycles. Data in accordance with 6.2.1.4 shall be recorded for each cycle.

4.6.16 Fungus test. If the valve contains fungus nutrients, the valve shall be subjected to a Fungus Test in accordance with Method 508, Procedure I, of MIL-STD-810. If no fungus nutrients exist in a valve, a statement to this effect shall be in the test report.

4.6.17 Salt fog test. The valve shall be subjected to a 50-hour Salt Fog Test in accordance with Method 509, Procedure I, of MIL-STD-810, while mounted in a manner similar to its intended installation. The valve shall be operated before and after the test period as specified in MIL-STD-810 and once every 5 hours during the test. Rated inlet conditions shall be used for all cycles. Data in accordance with 6.2.1.4 shall be recorded for each cycle.

4.6.18 Humidity test. The valve, when mounted in a manner similar to its intended installation, shall be subjected to a 25-hour Humidity Test. The humidity shall be maintained at 95 percent and the temperature at 51.7° ±14°C (125° ±25°F). The valve shall be operated before and after the test as specified in MIL-STD-810 and at the end of 5, 10, 20, and 25 hours. At the end of 15 hours, the test shall be interrupted and the valve shall be subjected to an ambient temperature of -53.9° ±2.8°C (-65° ±5°F) for 5 hours without drying, shaking, or otherwise removing condensed water from the valve. At the end of the 5-hour cold soak period, the valve shall be operated for one cycle. Rated inlet conditions shall be used for all cycles. Data in accordance with 6.2.1.4 shall be recorded for each cycle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

4.6.19 High temperature test. The valve shall be soaked at the high non-operating (exposure) temperature for 48 hours. The temperature shall then be reduced to high operating temperature and, while being maintained at this temperature, the valve shall be subjected to 10 continuous cycles of 60 seconds open followed by 60 seconds closed. The valve shall then be subjected to 10 continuous cycles of 5 minutes open followed by 5 minutes closed. All cycles shall be conducted using the applicable maximum temperature inlet conditions of table I.

4.6.20 Pressure drop test. Loss in static pressure from inlet to outlet port of the valve shall be determined for a range of conditions so that a plot of

$$\frac{W\sqrt{T}}{P} \quad \text{versus} \quad \frac{\Delta P}{P}$$

on logarithmic (log-log) graph paper can be made where:

W = Actual weight flow

T = Temperature

P = Upstream pressure

ΔP = Pressure drop,

Pressure drop (Δ P) in excess of that specified in 3.9.3 shall be cause for rejection.

4.6.21 External load test. The valve shall be mounted by one of its connecting flanges and the external loads specified in the detail specification applied at the opposite flange. Bending moment loads shall be applied in both directions in each of two mutually perpendicular planes whose intersection falls on the centerline of the connecting flanges. The test shall be repeated with the valve mounted on the opposite flange. There shall be no evidence of damage or permanent deformation to the valve as a result of the applied loads.

4.6.22 Burst pressure test. The valve shall be subjected to a Burst Pressure Test as follows: With the valve in the closed position and the outlet port open to atmospheric pressure, apply an air pressure equal to the minimum burst pressure specified in 3.7.16, to the inlet port of the valve for a period of 10 minutes. Repeat the test, except the valve shall be in the open position and the outlet port shall be capped. The unit shall show no evidence of structural failure upon completion of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

4.6.23 Icing test. The valve flow controlling mechanism icing test shall be specified in the detail specification. In addition, the following water contamination test shall be conducted. The valve shall be cycled five times at room temperature with saturated air. The temperature shall then be reduced to -32°C (-25°F) and the valve cycled 50 times. The valve shall operate within the limits specified in the detail specification.

4.6.24 Explosion proof. The valve shall be subjected to an explosion atmosphere test in accordance with MIL-STD-810, Method 511, procedure I with the exception that pressure or flow through the valve is not required. The unit shall not create an explosion.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Materials and methods used in packaging the valve shall be suitable to insure protection of the article from damage or deterioration. Unless otherwise specified in the applicable contract or order, the valve shall be packaged in accordance with Method III of MIL-P-116. Individual containers shall be so constructed as to allow removal of the valve for inspection purposes without damaging the container or labels.

5.2 Packing. Valves packaged in accordance with 5.1 shall be packed in accordance with MIL-STD-794. Where more than one valve is to be packed within a shipping container, the valve shall be individually packaged within the shipping container. The shipment marking nomenclature shall be, Valve, Starter Control, Pneumatic, Aircraft Engine.

5.3 Marking. In addition to any special marking required, valve packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

5.4 Packing list. The contractor shall furnish a packing list with each valve.

6. NOTES

6.1 Intended use. The valves covered by this specification are intended for use in aircraft and missiles using gas turbine engines.

6.2 Typical submittal data. To obtain approval of the starter valve, applicable data in the following subparagraphs should be included in compliance with 3.2.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

6.2.1 Preproduction test data

6.2.1.1 General test data requirements

6.2.1.1.1 Temperature corrections. Performance characteristics shall not be corrected for ambient air temperature. Temperature correction factors shall be furnished in the preproduction test report.

6.2.1.1.2 Pressure corrections. Performance characteristics shall not be corrected for ambient pressure. Pressure correction factors shall be furnished in the Preproduction Test Report.

6.2.1.2 Preproduction data summary. A summary shall be prepared by the contractor of all preproduction testing accomplished. It shall include, as a minimum, the general data required in the following paragraphs and all specific data required in the particular test paragraphs. Where the length of traces makes their inclusion difficult, only the initial and final portions and any other portions of significant interest shall be included. Component test reports covering all component tests conducted on each type of component shall also be submitted. These reports shall be prepared in accordance with MIL-STD-831.

6.2.1.3 Limits. Specification limits shall be superimposed on the plots of curves made from test data collected in preproduction testing.

6.2.1.4 Cycle data. A preproduction data summary sheet shall be prepared for each preproduction test unit and shall include the requirements of figure 1 as a minimum.

6.2.1.5 Disposition of preproduction test data. Preproduction test data shall be retained by the testing agency for 2 years and furnished to the procuring activity or authorized representative upon request.

6.2.1.6 Preproduction test data analysis. An analysis of all preproduction test data shall be conducted by the contractor and shall be included in the Preproduction Test Report. Any unusual condition, malfunction, failure, or out of specification performance shall be explained in detail in the test report and the procuring activity shall be notified in writing as soon as any of the above conditions are evident.

6.2.2 Quality conformance test data. Performance data shall be obtained from tests conducted on valves submitted for acceptance and shall not be corrected for ambient air temperature or ambient pressure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

TEST DATA SUMMARY SHEET

Items under Section 1 shall be recorded for all tests. Items under Section 2 shall be recorded for all quality conformance tests. Item 3 shall be recorded for all valves.

1. a. Date and time of cycle.

   b. Valve model designation (if applicable).

   c. Valve serial number.

   d. Barometric pressure.

   e. Ambient air temperature.

   f. Type of pretest treatment or conditioning.

   g. Time in and out of conditioning.

   h. Opening and closing time at specified inlet conditions (seconds).

   i. Pneumatic inlet conditions.

   j. Regulated pressure, if applicable.

   k. Voltage and amperage.

2. a. Results of dielectric test.

   b. Results of proof pressure test.

   c. Internal leakage.

   d. Total leakage.

   e. When specified in the detail specification, high speed trace of pressure rise v.s. time with point of electrical actuation shown.

3. Actual weight of the unit.

Figure 1. Test data summary sheet

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

6.2.2.1 Quality conformance data summary sheet. A quality conformance data summary sheet shall be prepared in accordance with the applicable portions of figure 1 for each valve.

6.2.3 Drawings and diagrams. When specified (see 6.2), the contractor shall prepare the following drawings and diagrams for submission to the procuring activity with the model specification. In addition, reduced size copies of the same drawings and diagrams shall be incorporated in the body of the model specification.

a. Valve installation drawing. The drawing shall show valve center of gravity, all connecting fittings, clearance required for installation and removal, filter locations, and cleaning instructions, if applicable.

b. Electrical diagram. The electrical diagram shall show all electrical circuits up to the connecting points of the aircraft.

c. A complete cutaway or cross sectional drawing of the valve and all components.

d. Packing requirements and packing drawings for the valve.

e. All contractor specifications listed in the model specification.

f. Preproduction test method (if available).

g. Control system schematic diagram.

6.3 Definitions and symbols. The applicable definitions and symbols used herein and in the model specification are as specified below:

6.3.1 Procuring activity. The procuring activity is the Department of Defense activity which is responsible for the valve contract.

6.3.2 Proof pressure. Proof pressure is that internal pressure to which all high pressure valve parts are pressure tested to prove their structural integrity without causing damage.

6.3.3 Minimum burst pressure. Minimum burst pressure is the greatest applied internal pressure to which the valve can be subjected before ultimate strength of the valve is exceeded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

6.3.4 Transfer time. Transfer time is the time interval from the removal of the valve or component from one conditioning chamber to the insertion of the valve or component into another conditioning chamber, or the time interval from the removal of the valve or component from the conditioning chamber to the initiation of testing.

6.3.5 Rating. A rating is the valve of some characteristic of performance as specified in the model specification. The valve shall be rated at that condition at which its associated starter is rated unless otherwise specified by the procuring activity.

6.3.6 Model specification. The model specification referred to herein is defined as the valve vendor prepared document used to define a particular valve.

6.3.7 Detail specification. The detail specification referred to herein is defined as a military or aircraft manufacturer prepared document used to define the applicable design and performance parameters required for a particular valve application.

6.3.8 Symbols

SYMBOLS	QUANTITY
g	Gravitational Constant
T	Temperature
Wv	Total weight of the valve
P	Pressure

6.4 Limitation of demonstrations. Demonstrations required of a valve and its components for the purpose of establishing a service type approval or production acceptance will be limited to those tests specified herein as modified by the individual valve model specification. This limitation of demonstrations will not relieve the valve manufacturer of responsibility for fulfilling all model specification requirements.

6.5 Identification of change. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 11

Review activity:
Air Force - 71

Preparing activity:
Air Force - 11

Project No. 2995-F005

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

FOLD

4950/TZS
Wright-Patterson AFB, O 45433

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEPARTMENT OF THE AIR FORCE
DoD-318

U.S.MAIL

4950/TZS
Wright-Patterson AFB, Ohio 45433

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

C 1 6 0 6 1

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:33Z
Check the source to verify that this is the current version before use.